

Md Hasan Shahriar

PHD CANDIDATE, DEPARTMENT OF COMPUTER SCIENCE
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(U.S. Permanent Resident, authorized to work in the U.S.)

RESEARCH INTERESTS

My research spans cyber-physical systems (CPS), artificial intelligence (AI), and cybersecurity, with a focus on enabling secure and resilient autonomous systems. I develop cross-layer defense frameworks to protect the physical layer (*intrusion detection and cyber-resilient CPS*), the intelligence layer (*trustworthy and resilient AI*), and the *cross-layer ML infrastructure* that integrates them, ensuring reliable operation even in the presence of coordinated cyber-physical and AI-driven threats.

EDUCATION

PhD in Computer Science

VIRGINIA TECH

Jan 2021–May 2026 (Expected)

Arlington, Virginia, USA

- Dissertation: *Toward Trustworthy Cyber-physical Systems: Robust Machine Learning for Secure Sensing, Perception, and Control*
- Advisor: Dr. Wenjing Lou

MS in Computer Engineering

FLORIDA INTERNATIONAL UNIVERSITY

Jan 2019–Dec 2020

Miami, Florida, USA

- Thesis: *Deception Defense against Stealthy Attacks in Power Grids*
- Advisor: Dr. Mohammad Ashiqur Rahman

BSc in Electrical and Electronic Engineering

BANGLADESH UNIVERSITY OF ENGINEERING AND TECHNOLOGY

Feb 2011–Mar 2016

Dhaka, Bangladesh

- Thesis: *Transient Stability Analysis of Smart Grids with Impacts of Distributed Generation*
- Advisor: Dr. Md Forkan Uddin

AWARDS, FELLOWSHIPS, & GRANTS

[A11] **Amazon Fellowship**, Amazon-VT Initiative for Efficient and Robust Machine Learning, 2024-2025.

[A10] **Student Travel Grant**, Maritime Cybersecurity Hackathon, 2025.

[A9] **Student Travel Grant for Attending IEEE ICDCS**, U.S. National Science Foundation, 2024.

[A8] **Student Travel Grant**, CyberTruck Challenge, 2024.

[A7] **Best Paper Runner Up Award**, Symposium on Vehicle Security and Privacy (VehicleSec), 2023.

[A6] **Student Travel Grant**, Inaugural Symposium on Vehicle Security and Privacy (VehicleSec), 2023.

[A5] **Fellowship for Graduate Student First-Author Papers**, Graduate School, Virginia Tech, 2023.

[A4] **Bangladesh-Sweden Trust Fund Scholarship**, July 2021.

[A3] **Student Travel Grant for Attending ACM WiSec**, U.S. Army Research Office, 2019.

[A2] **Admission Test Excellency Scholarship**, Bangladesh University of Engineering and Technology, 2011.

[A1] **Education Board Scholarship**, Government of Bangladesh, 2008 & 2010.

PUBLICATIONS

Journal Articles

- [J3] **Md Hasan Shahriar**, Mohammad Raashid Ansari, M. S. Haque, Jean-Philippe Monteuiis, Cong Chen, Jonathan Petit, Y. Thomas Hou, Wenjing Lou. “VehiGAN: Generative Adversarial Networks for Adversarially Robust V2X Misbehavior Detection Systems.” In *ACM Transactions on Cyber-Physical Systems (ACM TCPS)*, 2025. (Impact Factor: 2.0).
- [J2] **Md Hasan Shahriar**, Y. Xiao, P. Moriano, Wenjing Lou, Y. Thomas Hou. “CANShield: Deep Learning-Based Intrusion Detection Framework for Controller Area Networks at the Signal-Level.” In *IEEE Internet of Things Journal (IEEE IoT-J)*, 2023. (Impact Factor: 10.6).
- [J1] **Md Hasan Shahriar**, M. A. Rahman, M. Jafari, S. Paudyal. “Formal Analytics for Stealthy Attacks against Contingency Analysis in Power Grids.” In *Sustainable Energy, Grids and Networks (SEGAN)*, 2024. (Impact Factor: 5.6).

Conference Papers (Selected)

- [C15] **Md Hasan Shahriar**, Mohaimin Al Barat, Harshavardhan Sundar, Ning Zhang, Naren Ramakrishnan, Y. Thomas Hou, Wenjing Lou. “Detecting Temporal Misalignment Attacks in Multimodal Fusion for Autonomous Driving” Accepted to appear in *The International Conference on Learning Representations (ICLR)*, 2026. (Acceptance rate: 28%).
- [C14] **Md Hasan Shahriar**, Mohaimin Al Barat, Harshavardhan Sundar, Ning Zhang, Naren Ramakrishnan, Y. Thomas Hou, Wenjing Lou. “Temporal Misalignment Attacks against Multimodal Perception in Autonomous Driving” Accepted to appear in *IEEE Conference on Secure and Trustworthy Machine Learning (SaTML)*, 2026. (Acceptance rate: 26%).
- [C13] Kuan Yu Chen, **Md Hasan Shahriar**, Wen Wei Li, Shi Cho Cha, Wenjing Lou. “HOTWIRE: Real-World Impersonation and Discharge Attacks on Electric Vehicle Charging Systems” Accepted to appear in *USENIX WOOT Conference on Offensive Technologies (WOOT)*, 2026.
- [C12] **Md Hasan Shahriar**, Ning Wang, Naren Ramakrishnan, Y. Thomas Hou, Wenjing Lou. “Let the Noise Speak: Harnessing Noise for a Unified Defense Against Adversarial and Backdoor Attacks.” In Proceedings of *European Symposium on Research in Computer Security (ESORICS)*, 2025. (Acceptance rate: 17%).
- [C11] **Md Hasan Shahriar**, Mohammad Raashid Ansari, Jean-Philippe Monteuiis, Cong Chen, Jonathan Petit, Y. Thomas Hou, Wenjing Lou. “VehiGAN: Generative Adversarial Networks for Adversarially Robust V2X Misbehavior Detection Systems.” In Proceedings of *IEEE International Conference on Distributed Computing Systems (ICDCS)*, 2024. (Acceptance rate: 21%).
- [C10] **Md Hasan Shahriar**, Wenjing Lou, Y. Thomas Hou. “CANtropy: Time Series Feature Extraction-Based Intrusion Detection Systems for Controller Area Networks.” In Proceedings of *Symposium on Vehicle Security and Privacy (VehicleSec)*, 2023. **Best Paper Runner-Up Award**. (Acceptance rate: 36.0%)
- [C9] S. Shi, Y. Xiao, C. Du, **Md Hasan Shahriar**, A. Li, Ning Zhang, Y. Thomas Hou, Wenjing Lou. “MS-PTP: Protecting Network Timing from Byzantine Attacks.” In Proceedings of *ACM Conference on Security and Privacy in Wireless and Mobile Networks (WiSec)*, 2023. (Acceptance rate: 25.4%)
- [C8] **Md Hasan Shahriar**, Y. Xiao, P. Moriano, Wenjing Lou, Y. Thomas Hou. “CANShield: Signal-based Intrusion Detection for Controller Area Networks.” In *Embedded Security in Cars (ESCAR)*, 2022.
- [C7] **Md Hasan Shahriar**, Mohammad Ashiqur Rahman, Nur Imtiazul Haque, Badrul Chowdhury, S. G. Whisenant. “iDDAF: An Intelligent Deceptive Data Acquisition Framework for Secure Cyber-Physical Systems.” In Proceedings of *EAI International Conference on Security and Privacy in Communication Networks (SecureComm)*, 2021. (Acceptance rate: 34%)
- [C6] **Md Hasan Shahriar**, Mohammad Ashiqur Rahman, Nur Imtiazul Haque, Badrul Chowdhury. “DDAF: Deceptive Data Acquisition Framework against Stealthy Attacks in Cyber-Physical Systems.” In Proceedings of *IEEE 45th International Conference on Software Engineering (COMPSAC)*, 2021. (Acceptance rate: 27%)
- [C5] **Md Hasan Shahriar**, Alvi Ataur Khalil, Mohammad Ashiqur Rahman, Mohammad Hossein Manshaei, Dong Chen. “iAttackGen: Generative Synthesis of False Data Injection Attacks in Cyber-Physical Systems.” In Proceedings of *IEEE Conference on Communications and Network Security (CNS)*, 2021. (Acceptance rate: 26%)

- [C4] M. Jafari, **Md Hasan Shahriar**, Mohammad Ashiqur Rahman, S. Paudyal. “False Relay Operation Attacks in Power Systems with High Renewables.” In Proceedings of *IEEE Power & Energy Society General Meeting (PESGM)*, 2021.
- [C3] Nur Imtiazul Haque, **Md Hasan Shahriar**, Md Golam Dastgir, Anjan Debnath, Imtiaz Parvez, Arif Sarwat, and Mohammad Ashiqur Rahman. “Machine Learning in Generation, Detection, and Mitigation of Cyberattacks in Smart Grid: A Survey.” In Proceedings of *North American Power Symposium (NAPS)*, 2021.
- [C2] **Md Hasan Shahriar**, Nur Imtiazul Haque, Mohammad Ashiqur Rahman, Miguel Alonso Jr. “G-IDS: Generative Adversarial Networks Assisted Intrusion Detection System.” In Proceedings of *IEEE 45th International Conference on Software Engineering (COMPSAC)*, 2020. (Acceptance rate: 24%)
- [C1] **Md Hasan Shahriar**, Md Jawwad Sadiq, and Md Forkan Uddin. “Stability Analysis of Grid-connected PV Array Under Maximum Power Point Tracking”. In *Proceedings of International Conference on Electrical and Computer Engineering (ICECE)*, 2016.

Under Review (Ongoing)

- [O1] **Md Hasan Shahriar**, Ning Wang, Amit Kumar Sikder, Naren Ramakrishnan, Y. Thomas Hou, Wenjing Lou. “Noise, Why Can’t You Bend? Detecting Adversarial Perturbations in Wireless Sensing via Structural Fragility”

Theses

- [T3] **Md Hasan Shahriar**. “Toward Trustworthy Cyber-physical Systems: Robust Machine Learning for Secure Sensing, Perception, and Control” In *Virginia Tech Theses and Dissertations*, 2026 (Anticipated).
- [T2] **Md Hasan Shahriar**. “Deception Defense against Stealthy Attacks in Power Grids.” In *Florida International University Theses and Dissertations*, 2020.
- [T1] **Md Hasan Shahriar**. “Transient Stability Analysis of Smart Grids with Impacts of Distributed Generation.” In *Bangladesh University of Engineering and Technology Theses and Dissertations*, 2016.

PRESENTATIONS & TALKS

Invited Research Talks:

- “Temporal Misalignment Attack against Multimodal Fusion in Autonomous Driving”
4th Workshop on Future Automotive Research Datasets, November 2025.
- “Security of Connected and Autonomous Vehicles: From In-vehicular Networks to Multimodal Fusion”
Amazon VT Initiative Kickoff, Invited Talk, Blacksburg, VA, Fall 2024.
- “Generating State-of-the-art V2X Misbehavior Detection Dataset and a Robust Detection Approach”
3rd Workshop on Future Automotive Research Datasets, April 2024.
- “CANShield: Signal-based Intrusion Detection for Controller Area Networks”,
1st Workshop on Future Automotive Research Datasets, April 2021 and
ACIC-DoD ROLLAGE TEM, November 2021.
- “A Survey on CAN Intrusion Detection Dataset”
1st Workshop on Automotive Research Datasets, November 2021.
- “Deception-based Defense against False Data Injection Attacks in Power Grids”
CAPER Meeting (Virtual), Fall 2020.

Paper Presentations:

- **ESORICS 2025**, “Let the Noise Speak: Harnessing Noise for a Unified Defense Against Adversarial and Backdoor Attacks.”, September 2025.
- **IEEE ICDCS 2024**, “VehiGAN: Generative Adversarial Networks for Adversarially Robust V2X Misbehavior Detection Systems”, July 2024.
- **VehicleSec 2023**, “CANtropy: Time Series Feature Extraction-Based Intrusion Detection Systems for Controller Area Networks”, February 2023.
- **ESCAR USA 2022**, “CANShield: Signal-based Intrusion Detection for Controller Area Networks”, June 2022.

- **IEEE COMPSAC 2021**, “DDAF: Deceptive Data Acquisition Framework against Stealthy Attacks in Cyber-Physical Systems”, July 2021.
- **EAI SecureComm 2021**, “iDDAF: An Intelligent Deceptive Data Acquisition Framework for Secure Cyber-physical Systems”, September 2021.

Poster Presentations:

- **Amazon-VT’24**, “VehiGAN: Generative Adversarial Networks for Adversarially Robust V2X Misbehavior Detection Systems,” M. H. Shahriar, M. R. Ansari, J.-P. Monteuiis, C. Chen, J. Petit, Y. T. Hou, W. Lou, Fall Kickoff Meeting of Amazon-VT Initiative, Blacksburg, VA, 2024.
- **VehicleSec’23**, “CANtropy: Time Series Feature Extraction-Based Intrusion Detection Systems for Controller Area Networks,” M. H. Shahriar, W. Lou, Y. T. Hou. Symposium on Vehicle Security and Privacy (VehicleSec), 2023.
- **WiSec’19 & FICS’19**, “Poster: False Data Injection Attacks against Contingency Analysis in Power Grids,” M. Rahman, M.H. Shahriar, R. Masum, ACM Conference on Security and Privacy in Wireless and Mobile Networks (WiSec), 2019 & also in FICS Research Annual Conference on Cybersecurity, University of Florida, 2019.

RESEARCH APPOINTMENTS

- [R3] **Graduate Student Researcher** **2021 – Present**
 COMPLEX NETWORK AND SECURITY RESEARCH (CNSR) LAB, VIRGINIA TECH *Arlington, Virginia, USA*
Advisor: *Dr. Wenjing Lou*
 My Ph.D. research integrates cybersecurity, machine learning, and CPS to develop robust, resilient, and trustworthy AI for connected and autonomous vehicles. I designed intrusion detection systems for CAN and V2X networks to counter stealthy and adversarial attacks [C8,C10,C11,J2,J3], investigated network-induced and multimodal fusion attacks on autonomous perception [C13,O3], and developed a unified defense framework that mitigates both adversarial and backdoor ML threats across diverse modalities [C12,O2]. In addition, I mentored several graduate students whose work contributed to peer-reviewed and ongoing publications.
- [R2] **Graduate Student Researcher** **2019–2020**
 ANALYTICS FOR CYBER DEFENSE (ACyD) LAB, FLORIDA INTERNATIONAL UNIVERSITY *Miami, Florida, USA*
Advisor: *Dr. Mohammad Ashiqur Rahman*
 My M.S. research focused on securing cyber-physical systems, particularly detecting and mitigating stealthy threats in the smart grid. I developed a threat synthesizer combining formal methods and GANs to model complex attack behaviors [J1, C4, C5]. Building on this, I proposed a deception-based moving target defense to counter stealthy intrusions [C6, C7] and explored GAN-based defense models for network security [C2], while mentoring several undergraduate researchers.
- [R1] **Undergraduate Student Researcher** **2014–2016**
 BANGLADESH UNIVERSITY OF ENGINEERING AND TECHNOLOGY *Dhaka, Bangladesh*
Advisor: *Dr. Forkan Uddin*
 My undergraduate research focused on analyzing the robustness of different Maximum Power Point Tracking algorithms for the smart grid under cyberattacks and developing strategies to improve system resilience [C1].

INDUSTRIAL EXPERIENCE

- [I3] **Interim Engineering Intern** **May 2023–Aug 2023**
 QUALCOMM INCORPORATED *San Diego, California, USA*
Manager: *Jonathan Petit*, **Mentor:** *Jean-Philippe Monteuiis*
 - Executed real-world adversarial attacks on traffic sign detection systems to quantify adversarial robustness.
 - Investigated the transferability of adversarial examples across diverse object detection models.
 - Developed DVC-based pipelines for systematic dataset management and reproducible experiments.
- [I2] **Interim Engineering Intern** **May 2022–Aug 2022**
 QUALCOMM INCORPORATED *Boxborough, Massachusetts, USA*
Manager: *Jonathan Petit*, **Mentor:** *Rashed Ansari*
 - Researched and evaluated generative AI models (GANs) to synthesize realistic yet fake V2X messages.
 - Designed a GAN-based misbehavior detection to effectively detect anomalous basic safety messages (BSMs).
 - Continued the collaboration beyond internship and extended this project, which resulted in [C11,J3].

[I1] Assistant Engineer (Electrical)

ELECTRICITY GENERATION COMPANY BANGLADESH LTD.

Sep 2017 – Dec 2018*Dhaka, Bangladesh*

- Operated a 2x120 MW gas turbine power plant by coordinating with the national load dispatch center.
- Developed operational and maintenance schedules to minimize downtime through proactive planning.

TEACHING EXPERIENCE**Guest Lecturer, Department of Computer Science**

VIRGINIA TECH

Guest lectured for the following graduate course:

Spring 2026*Blacksburg, VA*

- **CS 5984: Fundamentals of Information Security**

Spring 2026

Lecturer, Department of Computer Science

UTTARA UNIVERSITY

Taught the following undergraduate courses and led corresponding lab sessions:

May 2016 – May 2017*Uttara, Dhaka, Bangladesh*

- **EE 101: Electrical Circuits**
- **EE 205: Basic Electronics**
- **EE 210: Digital Logic Design**
- **EE 315: Microprocessor Interfacing and Low-Level Language**

Fall 2016

Spring 2017

Fall 2016, Spring 2017

Fall 2016, Spring 2017

STUDENT MENTORSHIP

- **Kuan Yu Chen (MS, National Taiwan University of Science and Technology):** EV Charging Security.
- **Md Mohaimin Al Barat (PhD, Virginia Tech):** Security of Multimodal Fusion in Autonomous Driving.
- **Sydney Johns (PhD, Virginia Tech):** Practical Intrusion Detection Systems for Cyber-physical Systems.
- **Md Shahedul Haque (MS, Virginia Tech):** Defacing Technique in MRI data for Privacy Preserving ML.
- **Samara Ruiz Sandoval (Undergrad, Florida International University):** ML for Security of Smart Grid.

FEATURED

Amazon-Virginia Tech Initiative awards two student fellowships, five faculty research awards, VT News, 10/22/2024

PROFESSIONAL SERVICES**Artifact Evaluation Committee:**

- ACM Conference on Computer and Communications Security (CCS) 2025
- USENIX WOOT Conference on Offensive Technologies (WOOT) 2026

Journal Reviewer:

- IEEE Internet of Things Journal (IoT-J) 2025
- IEEE Transactions on Big Data (TBD) 2025
- ACM Transactions on Cyber-Physical Systems (TCPS) 2025
- IEEE Transactions on Vehicular Technology (TVT) 2024
- IEEE Transactions on Computers (TC) 2024
- Computers & Security (C&S) 2024
- IEEE Sensors Journal (SJ) 2023
- IEEE Transactions on Information Forensics and Security (TIFS) 2023
- Vehicular Communications (VehiCom) 2023
- IEEE Power & Energy Society Transactions on Power Systems (IEEE PES) 2021
- International Journal of Electronic Security and Digital Forensics (IJESDF) 2021

External Conference Reviewer:

- ACM ASIA Conference on Computer and Communications Security (AsiaCCS) 2025
- IEEE Symposium on Security and Privacy (IEEE S&P) 2022–2025
- European Symposium on Research in Computer Security (ESORICS) 2022–2024
- ACM Conference on Security and Privacy in Wireless and Mobile Networks (ACM WiSec) 2022–2025

- IEEE Conference on Communications and Network Security (IEEE CNS) 2022–2024
- International Conference on Computer Communication and Networks (ICCCN) 2023
- IEEE International Conference on Distributed Computing Systems (ICDCS) 2022
- IEEE International Conference on Communications (ICC) 2020
- International Symposium on Network Systems Security (NSysS) 2019

Community and Outreach Involvement:

- Student Volunteer, IEEE International Conference on Distributed Computing Systems (ICDCS), 2024
- Campus Representative, Graduate Student Assembly (GSA) – DC Region, Virginia Tech, 2023
- Session Chair, IEOM North American Industrial Engineering and Operations Management Conference, 2022
- Judge, Engineering Section, Northern Virginia Regional Science Fair, 2022
- Student Volunteer, ACM Conference on Security & Privacy in Wireless and Mobile Networks (WiSec), 2020

GOOGLE SCHOLAR

h-index: 8, i10-index: 8, citations: 429 (as of 5 October, 2025)

Link to Google Scholar : <https://scholar.google.com/citations?user=TcCzjTQAAAAJ&h>

REFERENCES

Dr. Wenjing Lou

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W. C. English Endowed Professor
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Dr. Y. Thomas Hou

IEEE Fellow
Bradley Distinguished Professor
Department of Electrical and Computer Engineering
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2040-K Torgersen Hall, Blacksburg, VA 24061
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Dr. Naren Ramakrishnan

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